

3.1 Waves + electromagnetic spectrum — 5054 Paper 2

Newest → oldest. Kept subquestions only. Mark scheme is those parts only, not the rest of the question.

2025

Oct/Nov 5054/22

Question 5 • included (a), (b)

EM map + $v = f\lambda$

(a) Fig. 5.1 represents the electromagnetic spectrum divided into the seven main regions.

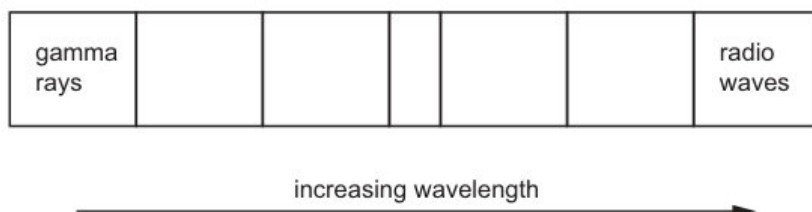


Fig. 5.1

The regions are arranged from left to right in order of increasing wavelength.

On Fig. 5.1:

- indicate the microwave region by marking it with the letter M
- indicate the ultraviolet region by marking it with the letters UV.

[1]

(b) The frequency of the microwaves used by some satellite television systems is 12 GHz.

Mark scheme — only (a), (b)

5(a)		B1
5(b)(i)	$(\lambda =) \frac{c}{f} \text{ or } \frac{3.0 \times 10^8}{1.2 \times 10^9}$ $2.5 \times 10^{-2} \text{ (m)}$ 0.025 (m)	C1 C1 A1
5(b)(ii)	<u>encoded microwave / microwave signal / microwave carrying data / microwave transmitting data</u> (to satellite from Earth) satellite transmits microwave to Earth / dish / antenna / subscriber / television	B1 B1

May/June 5054/22

Question 4 • included (a), (b), (c)

(a) Define 'power'.

.....
 [1]

(b) The efficiency of the motor is 70%.

Calculate the input power to the motor.

input power =W [2]

(c) The fork-lift truck lifts a load with mass of 50 kg through a vertical distance of 2.3 m.

Mark scheme — only (a), (b), (c)

4(a)	<i>motion of particles:</i> backwards and forwards / to and fro / closer and further apart	B1
	(oscillation) in the direction of travel of the wave / to the right / in direction of microphone / longitudinal / horizontal / parallel to vibrations of tuning fork or mention of compressions and rarefactions	B1
4(b)(i)	number of oscillations per second or number of wave(length)s (passing a point) per second	B1
4(b)(ii)	2.5 (waves) seen or (1 wave takes) 0.02 (s) or ($f = 1 / T$ or (number of) waves / time	C1
	50 (Hz)	A1
4(b)(iii)	quieter / less volume	B1
4(b)(iv)	half frequency shown and at least one complete oscillation seen	B1
4(c)(i)	(at least) three correctly curved wavefronts centered on gap	B1
	same wavelength as on left and some curving in correct direction	B1
4(c)(ii)	diffraction	B1

May/June 5054/21

Question 5 · included (a), (b), (c)

- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.

- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.

Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed

wavelength [2]

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.



Fig. 5.1

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]

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5054/21/M/J/25

[Turn over]

Mark scheme — only (a), (b), (c)

5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	<i>speed</i> : no change	B1
	<i>wavelength</i> : decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1

2024

Oct/Nov 5054/21

Question 5 · included (b), (c)

not vacuum

(b) Describe how a longitudinal wave differs from a transverse wave.

.....

.....

..... [2]

(c) An ultrasound transmitter used for a medical scan produces an ultrasound wave of frequency 8.4 MHz.

In soft human tissue, ultrasound travels at 1500 m/s.

Mark scheme — only (b), (c)

5(b)	(in a longitudinal wave,) vibration (of particles) is parallel to the propagation direction	B1
	in a transverse wave, vibration (of particles) is perpendicular to the propagation direction	B1
5(c)(i)	$(\lambda =) v / f$ or $1500 / 8.4 \times 10^6$	C1
	1.8×10^4	C1
	1.8×10^{-4} (m)	A1
5(c)(ii)	frequency: does not change	B1
	wavelength: increases	B1

Question 4 · included (a), (b), (c), (d)

visible + IR

(a) (i) Calculate the initial gravitational potential energy of the ball.

gravitational potential energy = J [2]

- (b) When the ball hits the ground, it stops moving.

The kinetic energy of the ball is transferred to internal energy.

The temperature of the ball increases.

- (i) The specific heat capacity of modelling clay is $1400 \text{ J/(kg } ^\circ\text{C)}$.

Calculate the maximum possible temperature increase of the ball. Show your working.

temperature increase = $^\circ\text{C}$ [2]

- (ii) Suggest **one** reason why the actual temperature increase of the ball may be smaller than the value calculated in (b)(i).

.....
.....
..... [1]

[Total: 8]

- (c) The gas in the cylinder reaches a temperature of 100°C .

The piston is released and moves to the right. The temperature remains at 100°C .

The pressure of the gas now decreases.

- (i) Explain, in terms of the particles of the gas, why the pressure of the gas now decreases.

.....
.....
..... [1]

- (ii) Eventually, the piston stops moving.

Explain why.

.....
.....
.....
..... [2]

[Total: 10]

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S MARGIN

(d) On Fig. 4.1, point P is shown immediately next to point R.

No light is observed at point P, but a detector at point P registers radiation reaching the detector.

State which region of the electromagnetic spectrum is detected at point P.

..... [1]

[Total: 8]

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5054/21/O/N/24

DO NOT WRITE IN THIS MARGIN

Mark scheme — only (a), (b), (c), (d)

4(a)	at least one ray from left-hand side showing correct refraction at both faces and no incorrect rays	B1
	at least two correct rays (i.e. two rays within glass) and one ending at R and the other at V	B1
4(b)	both (red and violet light) travel more slowly in glass than in air	B1
	red light travels more quickly (than violet light) in glass	B1
4(c)(i)	any four (of the five) colours in order or five colours in any order	C1
	red orange yellow green blue indigo violet	B1

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4(c)(ii)	frequency increases and wavelength decreases	B1
4(d)	infrared (radiation)	B1
Question	Answer	Marks

May/June 5054/21

Question 4 · included (a)(i), (a)(ii), (b), (c)

not ultrasound use

(b) Sound waves are longitudinal waves. Water waves are transverse waves.

- (i)** Describe, by referring to the movement of the particles in the wave, the difference between a longitudinal wave and a transverse wave.

You may include a labelled diagram to help your description.

.....
.....
.....

- (i)** State what is meant by 'frequency'.

.....
..... [1]

- (ii)** Calculate the speed of ultrasound in air.

Show your working.

speed = m/s [2]

(c) Fig. 4.1 shows radio waves of long and short wavelength passing over the same hill.

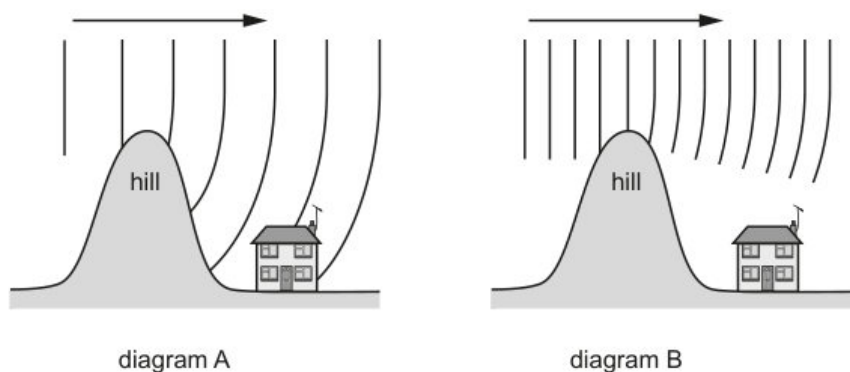


Fig. 4.1 (not to scale)

Explain why the radio waves in diagram A reach the house but the radio waves in diagram B do not reach the house.

.....

.....

Mark scheme — only (a)(i), (a)(ii), (b), (c)

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Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 [Guidance for chemical equations](#)

2023

Oct/Nov 5054/22

Question 10 · included (b)

only $v=f\lambda$

- (b) The gamma radiation produced has a frequency of 8.8×10^{19} Hz.

The speed of electromagnetic radiation in a vacuum is 3.0×10^8 m/s.

Calculate the wavelength of this gamma radiation in a vacuum.

wavelength = m [2]

[Total: 9]

Mark scheme — only (b)

10(b)	$(\lambda =) v/f = 3.0 \times 10^8 / 8.8 \times 10^{19}$	C1
	3.4×10^{-12} (m)	A1

Question 6 · included (a)

frequency

- (a) State what is meant by 'frequency'.

.....
 [2]

Mark scheme — only (a)

6(a)	(frequency =) the <u>number</u> of wavelengths	B1
	(passing a point) per <u>unit</u> time	B1

Oct/Nov 5054/21

Question 5 · included (a), (b)

EM + trans/long

- (b) Electromagnetic radiation is a transverse wave.

Describe the difference between a transverse wave and a longitudinal wave.

.....

.....

..... [2]

[Total: 6]

- (a) The name of one of these three regions is the ultraviolet region.

Mark scheme — only (a), (b)

5(a)(i)	infrared (radiation) and visible (light) and ultraviolet (radiation)	B1
	infrared (radiation) and visible (light) and ultraviolet (radiation) and in this order	B1
5(a)(ii)	damaging effect: (skin) cancer / cataracts	B1
	property: it / ultraviolet radiation is ionising	B1

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5(b)	in a transverse wave, the <u>vibration</u> direction is perpendicular / at right angles to the propagation direction	B1
	in a longitudinal wave, the <u>vibration</u> direction is parallel to the propagation direction	B1

Question	Answer	Marks
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Question 4 · included (a), (b)(i)

not n or rays

- (i) State what happens to the speed, to the frequency and to the wavelength of the light as it enters the block.

speed

frequency

wavelength [2]

(a) Calculate the wavelength in air of this red light.

wavelength = m [3]

Mark scheme — only (a), (b)(i)

4(a)	$(\lambda =) v / f$ or $3.0 \times 10^8 / 4.7 \times 10^{14}$	C1
	6.4×10^N (m) or $3 / 4.7 \times 10^6$	C1
	6.4×10^{-7} (m)	A1
4(b)(i)	speed: decreases frequency: stays the same wavelength: decreases	B2
4(b)(ii)	$i = 45^\circ$ and $r = 30^\circ$	C1
	$(n =) \sin(i) / \sin(r)$ or $\sin(45^\circ) / \sin(30^\circ)$	C1
	1.4	A1
4(b)(iii)	straight continuation of ray in block and refraction away from normal in air	B1

May/June 5054/22

Question 7 · included (a)

UV / shorter λ

7 (a) Ultraviolet radiation is one component of the electromagnetic spectrum.

Mark scheme — only (a)

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7	<u>Guidance for chemical equations</u>

Question 5 · included (a), (b)

- (a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

- (b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

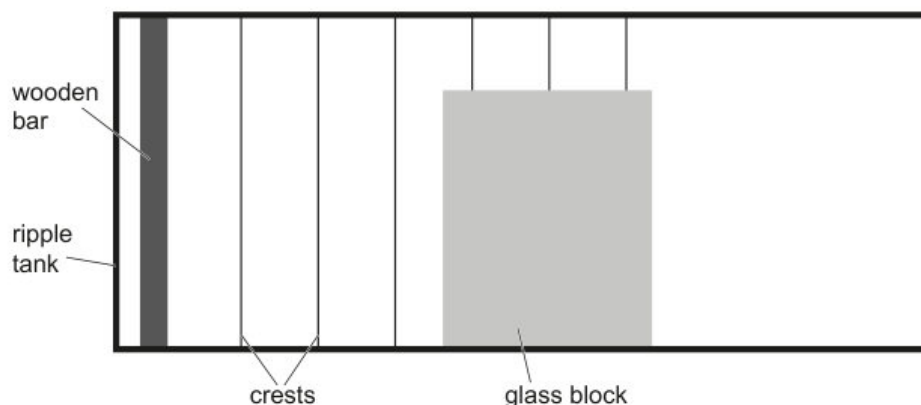


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

Mark scheme — only (a), (b)

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May/June 5054/21

Question 7 · included (a), (b), (c)

- (a) Describe the motion of the glider in the first 12 s.

.....

.....

..... [2]

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- (b) In the first 6.0 s of the motion, there is a resultant force of 1800 N on the glider.

Using the increase in speed in the first 6.0 s, calculate the mass of the glider.

mass = kg [3]

- (c) Determine the distance travelled by the glider in the first 6.0 s of its motion.

Mark scheme — only (a), (b), (c)

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7 [Guidance for chemical equations](#)

2022

Oct/Nov 5054/22

Question 4 · included (b), (c)

- (b) The pile-driver lifts the block from the top of a concrete cylinder, through a height of 0.80 m.

The gravitational field strength g is 10 N/kg.

(c) Describe what is meant by 'a rarefaction'.

.....
..... [1]

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Mark scheme — only (b), (c)

4(b)	(they) vibrate	M1
	(vibration) parallel to / in the direction of propagation / of energy travel	A1
4(c)	(point where) molecules are further apart (than average) or (point where) pressure / density is lower (than average)	B1

Oct/Nov 5054/21

Question 4 · included (a), (b), (c)

X-rays as EM

- (a) Calculate the change in gravitational potential energy (g.p.e.) of the car as it travels from A to B.

change in g.p.e. = [2]

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- (b) Calculate the speed of the car when it reaches B.

speed = [3]

- (c) After releasing it, the child expects the car to follow the track along the route ABCDEFG. In fact, the model car does not reach F.

Mark scheme — only (a), (b), (c)

4(a)	electromagnetic (waves) or components of electromagnetic spectrum	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(i)	number of oscillations / cycles / wavelengths	B1
	per unit time	B1
4(b)(ii)	gamma rays	B1
4(c)(i)	X-rays directed towards body part with (suspected) broken bone	B1
	X-rays pass through flesh but not (to same extent) through bones	B1
	image produced by digital / electronic / CCD / CMOS detector	B1
4(c)(ii)	X-rays (may) cause cancer / cell mutation / birth defect / miscarriage or X-rays show the bone structure rather than foetal development or image would not be clear	B1

May/June 5054/22

Question 8 · included (a), (b), (c)

not satellite

- 8 (a) Fig. 8.1 shows a ripple tank and the crests of the water wave that is produced in it.

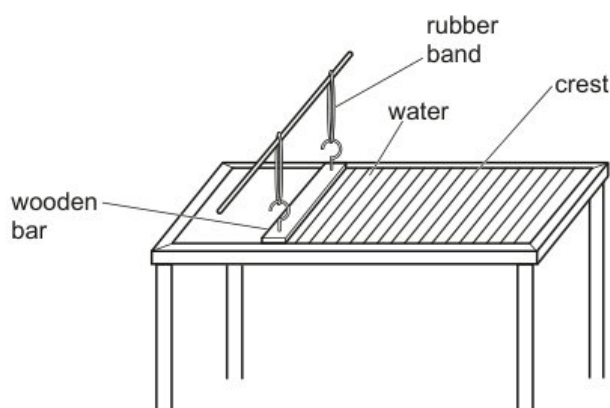


Fig. 8.1

The frequency of the water wave is 2.0 Hz and its amplitude is 3.0 mm.

(c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

(i) State the additional apparatus needed to demonstrate refraction.

..... [1]

(ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.



(b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
..... [1]

- (b) (i) The frequency of the wave is increased.

Describe how the apparatus shown in Fig. 8.1 is adjusted so that the frequency of the wave is increased.

.....
 [1]

- (ii) State what happens to the speed and wavelength of the wave as the frequency increases.

speed
 wavelength [2]

- (c) The apparatus shown in Fig. 8.1 can be used to demonstrate refraction.

- (i) State the additional apparatus needed to demonstrate refraction.

..... [1]

- (ii) Draw on Fig. 8.3 to show the refraction of the water wave.

Label a boundary where the refraction occurs.



Fig. 8.3

[3]

Mark scheme — only (a), (b), (c)

	at least two values thinking distance + speed (or inverse) calculated or all values of thinking distance/speed stated to be 0.45 (or inverse 2.2) or proportionality shown by at least two values or double the speed doubles the (thinking) distance or each 20 (km/h) / same increase in speed covers an extra 9 (m) distance	A1
7(b)(iii)	thinking distances increases	B1
	braking distances no change	B1

2021

Oct/Nov 5054/22

Question 3 · included (a)

- (a) Light travels more slowly in the block than in air.

Mark scheme — only (a)

Use official MS for the listed parts only.

Oct/Nov 5054/21

Question 8 · included (b)(iii), (b)(v)

- (b) An uncharged, conducting sphere is suspended from an insulating thread. The positively charged rod is placed near to the sphere, as shown in Fig. 5.1.

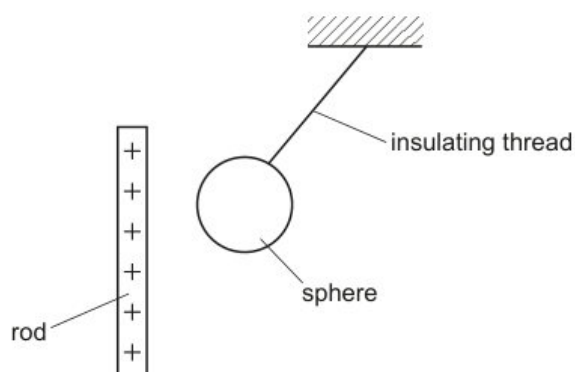


Fig. 5.1

- (i) On Fig. 5.1, draw the distribution of charge on the sphere.

[2]

Mark scheme — only (b)(iii), (b)(v)

Use official MS for the listed parts only.

Question 10 · included (a), (b)

- 10 (a) Fig. 10.1 shows a water wave moving from shallow into deep water. The wavefronts shown represent the crests of the wave.

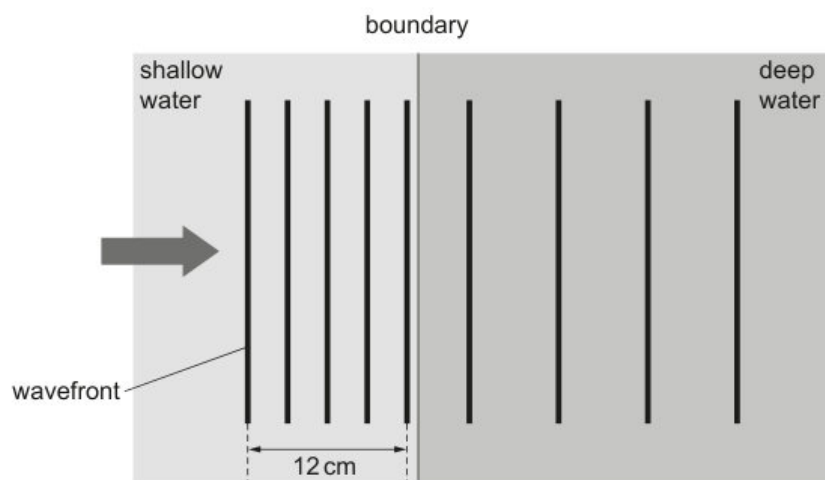


Fig. 10.1 (not to scale)

The water wave is made by dipping a wooden bar up and down in the water. The bar makes 10 complete up and down movements in 5.0 s.

- (b) White light is made up of different colours.

Mark scheme — only (a), (b)

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7 Guidance for chemical equations

Question 4 · included (a)

- 4 (a) Fig. 4.1 shows a wave on a rope and Fig. 4.2 shows a wave on a spring. Both waves are moving in the direction shown by the arrows.



Fig. 4.1



Fig. 4.2

Mark scheme — only (a)

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7 Guidance for chemical equations

2020

Oct/Nov 5054/21

Question 4 • included (a)

- (a) Light travels more slowly in glass than in air.

Mark scheme — only (a)

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7 Guidance for chemical equations

May/June 5054/22

Question 10 • included (b), (c)(i), (c)(ii)

- (b) Fig. 10.1 is the trace on the screen of the oscilloscope.

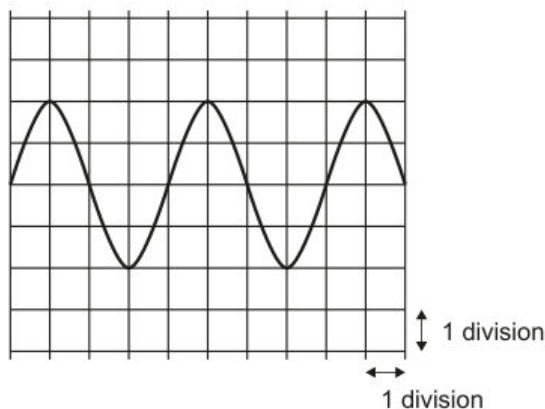


Fig. 10.1

The settings on the oscilloscope are 10 ms/division for the x-axis and 3.0 V/division for the y-axis.

- (c) The trace shown in Fig. 10.1 is caused by a sound.

The sound travels through the air to a microphone from the place that it is made. The microphone is connected to the oscilloscope which displays the waveform shown.

- (i) Sound is a type of wave.

State which type.

..... [1]

- (ii) Describe the motion of the air molecules as the sound passes through the air to the microphone.

.....

 [2]

Mark scheme — only (b), (c)(i), (c)(ii)

Use official MS for the listed parts only.

Question 5 · included (a)

EM uses

- (a) Draw **one** line from each component of the spectrum to another suitable use for that component.

component	use
	sunbeds
microwaves	television controller
	satellite television
ultraviolet	airport security check of cases

[2]

Mark scheme — only (a)

- should be treated as a single incorrect response
- Non-contradictory responses after the first *n* responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

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May/June 5054/21

Question 9 · included (a)(ii)

X-ray $v = f\lambda$

- (ii) Describe what happens to ultrasound waves as they meet the boundary between two different materials.

.....
..... [2]

Mark scheme — only (a)(ii)

Use official MS for the listed parts only.

2019

Oct/Nov 5054/22

Question 5 · included (a), (b)(i)

- (i) The speed of ultrasound in the fluid is 1500 m/s.

Calculate the wavelength of the ultrasound in the fluid.

wavelength = [2]

- (a) Describe what is meant by *longitudinal*.

.....
.....
..... [2]

Mark scheme — only (a), (b)(i)

Use official MS for the listed parts only.

Oct/Nov 5054/21

Question 10 · included (a), (b), (c), (d)

- (a) State the speed of light in a vacuum.

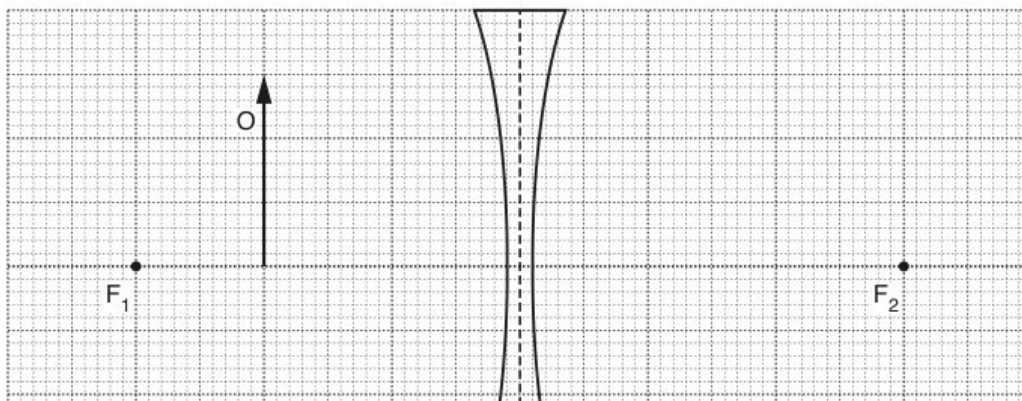
..... [1]

- (b) State the colours of the visible spectrum.

..... [1]

- (c) An object O of height 3.0 cm is placed 4.0 cm from the centre of a diverging lens.

Fig. 10.1 shows the object O, the diverging lens and the two focal points (principal focuses), F_1 and F_2 , of the lens.



- (d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

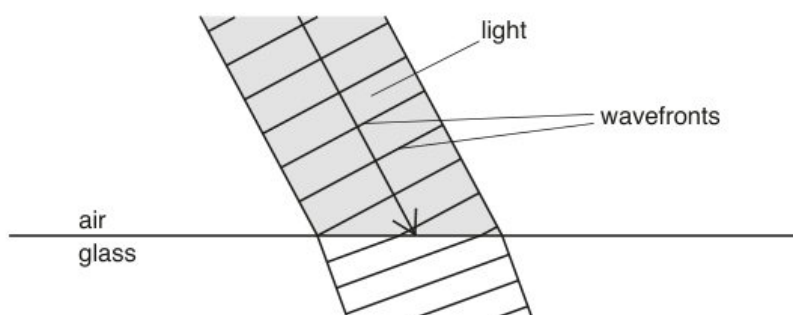


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....
.....

- (d) Light passing from air into glass refracts in a similar way to a water wave passing from deep water into shallow water.

Fig. 10.2 represents light passing from air into glass at an angle to the surface.

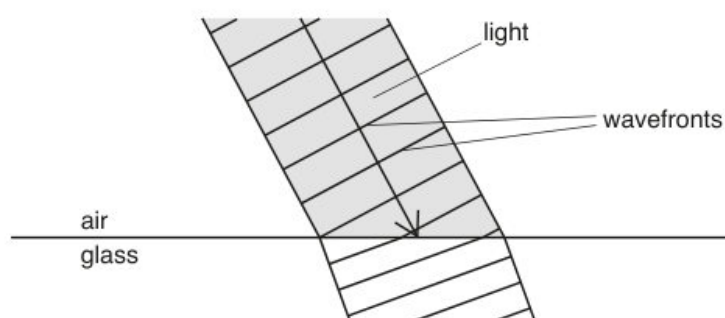


Fig. 10.2

One side of a wavefront strikes the glass before the other side.

Explain why the wavefronts change direction as the light enters the glass.

.....

.....

Mark scheme — only (a), (b), (c), (d)

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Question 4 · included (a), (b)

EM labels

- (a) State the name of the component labelled:

- A
- B
- C [2]

- (b) State the name of the component of the electromagnetic spectrum that has:

the greatest frequency

the smallest wavelength [1]

Mark scheme — only (a), (b)

Use official MS for the listed parts only.

May/June 5054/22

Question 4 · included (a), (b), (c)

star EM

- (a) Define the term *pressure*.

.....
..... [1]

- (b) The density of mercury is $1.4 \times 10^4 \text{ kg/m}^3$. The gravitational field strength g is 10 N/kg .

The pressure of the atmosphere is $1.0 \times 10^5 \text{ Pa}$.

Calculate the pressure of the gas in the container.

pressure = [3]

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5054/22/M/J/19

- (c) In Fig. 2.1, the mercury level on the left-hand side of the manometer is lower than on the right-hand side.

The gas inside the container is heated. This causes the mercury levels on both sides to become the same.

Mark scheme — only (a), (b), (c)

Use official MS for the listed parts only.

May/June 5054/21

Question 10 · included (a)

- (a) (i) Determine the wavelength of the wave in the ripple tank.

wavelength = [2]

Mark scheme — only (a)

Use official MS for the listed parts only.

2018

Oct/Nov 5054/22

Question 9 · included (b), (c)

- (b) Sound and ultrasound are longitudinal waves that consist of compressions and rarefactions.

(c) A sound wave travelling in a liquid, passes into air.

(i) State what happens to the speed of the sound wave as it enters the air.

..... [1]

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Mark scheme — only (b), (c)

Use official MS for the listed parts only.

Oct/Nov 5054/21

Question 9 · included (a), (b)

(a) State how a longitudinal wave differs from a transverse wave.

.....
..... [1]

(b) A vibrating rod produces a water wave in a ripple tank. Fig. 9.1 shows the crests of the wave passing into the right-hand section of the tank where the depth of the water is different from the depth in the rest of the tank.

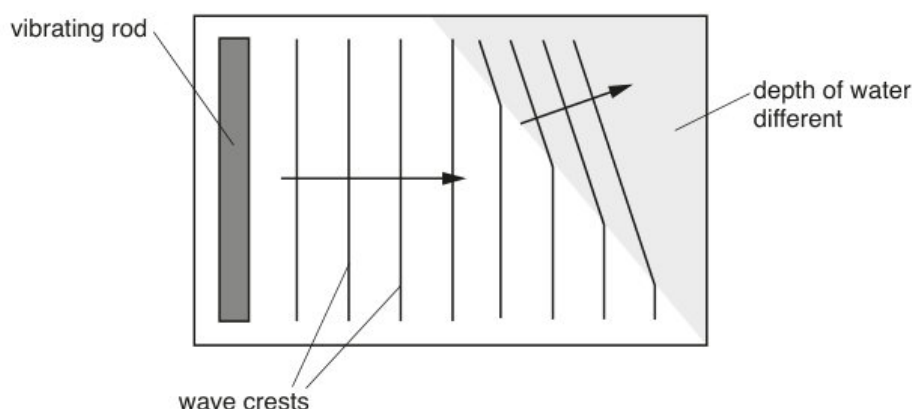


Fig. 9.1

The arrows on Fig. 9.1 show the direction of travel of the wave in the two sections of the ripple tank.

Mark scheme — only (a), (b)

Use official MS for the listed parts only.

May/June 5054/22

Question 5 · included (b)

(b) The object has a constant speed for some of the time.

Mark scheme — only (b)

Use official MS for the listed parts only.

May/June 5054/21

Question 11 · included (a), (b)

- (b) In one type of bathroom shower, cold water passes through a metal pipe which contains an electric heater.

The cold water is heated and emerges from the shower head.

The temperature of the cold water before heating is measured and the hot water emerging from the shower in 1.0 minute is collected in a container.

Measurements and other data are:

temperature of water before heating = 16°C

temperature of water after heating = 37°C

volume of water collected in 1.0 minute = $4.6 \times 10^{-3} \text{ m}^3$

specific heat capacity of water = $4200 \text{ J / (kg }^{\circ}\text{C)}$

density of water = 1000 kg / m^3

- (i) Calculate the mass of water leaving the shower in 1.0s.

mass =[2]

- 10 (a) (i) Describe two differences between boiling and evaporation.

1.
.....
.....
2.
.....
.....
- [4]

Mark scheme — only (a), (b)

Use official MS for the listed parts only.

Question 9 · included (a), (c)

EM T/F + speed

- (c) The Earth is $1.5 \times 10^8 \text{ km}$ from the Sun.

- (a) Some of the statements below are true and some of them are false.

Put a tick (✓) in the box after each statement to show whether it is true or false.

	true	false
Gamma rays are used to kill cancerous cells but can also cause cancer.	☐	☐
Infra-red is used in sun-beds.	☐	☐
Radio waves have the highest frequency in the electromagnetic spectrum.	☐	☐
The higher the frequency of the radiation the smaller is the wavelength in air.	☐	☐

[2]

Mark scheme — only (a), (c)

Use official MS for the listed parts only.

2017

Oct/Nov 5054/22

Question 6 · included (a), (b), (c)

- (a) State what is meant by the term *echo*.

.....
.....[1]

- (b) The pitch of the echo is the same as that of the original sound but the echo is not as loud.

State what has happened to

- (c) The speed of sound in air is 330 m/s.

Calculate the wavelength of this sound.

wavelength =[2]

Mark scheme — only (a), (b), (c)

Use official MS for the listed parts only.

Oct/Nov 5054/21

Question 10 · included (a), (b), (c)

$EM + v = f\lambda$

- (a) State the name of the component of the electromagnetic spectrum that is found in region
- (b) On Fig. 10.1, mark a tick (✓) in all of the boxes that represent the components with frequencies greater than the frequencies of visible light. [1]
- (c) The speed of electromagnetic radiation in a vacuum is $3.0 \times 10^8 \text{ m/s}$.

A television remote controller uses infra-red radiation of wavelength $9.4 \times 10^{-7} \text{ m}$.

- (i) Calculate the frequency of this radiation.

frequency = [3]

- (ii) Explain how infra-red radiation is used in television remote controllers.

.....

.....

.....

..... [2]

Mark scheme — only (a), (b), (c)

Use official MS for the listed parts only.

May/June 5054/22

Question 6 · included (a), (b)(i)

- (i) Calculate the wavelength of the sound.

wavelength = [2]

- 6 (a) Explain the difference between a longitudinal wave and a transverse wave.

You may draw a diagram, if you wish, to help your explanation.

.....

.....

.....

.....

.....[2]

Mark scheme — only (a), (b)(i)

Use official MS for the listed parts only.

May/June 5054/21

Question 10 · included (b), (c)(i), (c)(iii)

- (b) Fig. 10.2 shows wavefronts in a ripple tank. They move in the direction of the arrow.

The wave hits the boundary between two regions and the wave slows down as it enters the shaded region.

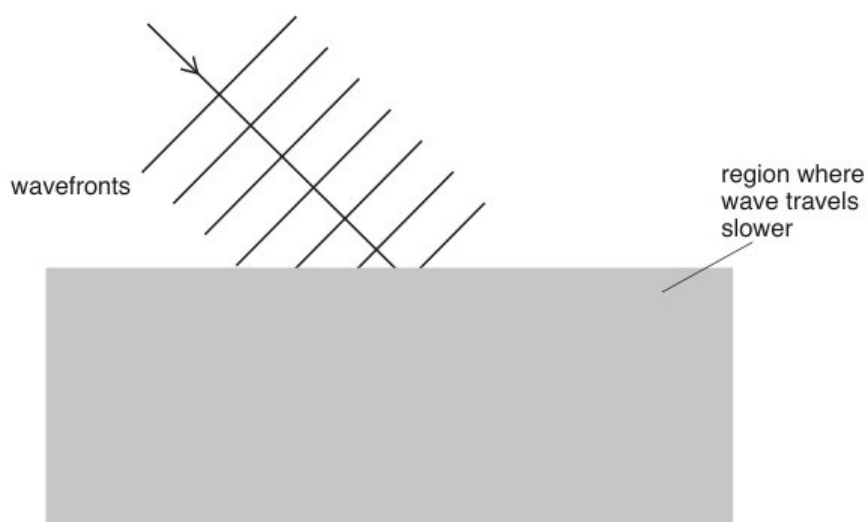


Fig. 10.2

- (i) State what is meant by *wavefront*.

- (i) Calculate the speed of sound in air.

speed =[2]

- (ii) 1. State the range of frequencies that can be heard by a healthy human ear.
.....[1]
2. Calculate the smallest wavelength of sound that can be heard by a healthy human ear.

wavelength =[1]

- (iii) Describe a simple experiment to show that sound waves obey the law of reflection.
You may draw a diagram if you wish.

Mark scheme — only (b), (c)(i), (c)(iii)

Use official MS for the listed parts only.

2016

Oct/Nov 5054/22

Question 10 · included (a)

- (a) Calculate the speed of the skier after 5.0 s.

speed =[2]

Mark scheme — only (a)

10 (a) (i)	$3.0 \times 10^8 \text{ m/s}$	B1
(ii)	$(\lambda =)c / f$ or $3.0 \times 10^8 / 4.3 \times 10^{14}$ $7.0 \times 10^{-7} \text{ m}$	C1 A1 [3]
(b) (i)	decreases	B1
(ii)	$\sin(i) = n \times \sin(r)$ or $1.5 \times \sin(30^\circ)$ or 0.75 49°	C1 A1
(iii)	41°	B1 [4]
(c) (i)	dispersion at both surfaces and refractions in correct direction	B1

Oct/Nov 5054/21

Question 5 · included (a), (b)

- (a) State what is meant by the *frequency* of a sound.

.....

[1]

- (b) (i) The speed of sound in air is 330 m/s.

Calculate the wavelength of the sound in the air.

wavelength =[2]

Mark scheme — only (a), (b)

- | | | |
|---------|---|----------|
| 5 (a) | number of oscillations / vibrations / wavelengths / compressions / rarefactions / cycles per second / unit time | B1 |
| (b) (i) | $(\lambda =)c/f$ or 330/2200
0.15 m | C1
A1 |
| (ii) | 1 no change
and
2 increases | B1 |
| (c) (i) | 1 loudspeaker vibrates / oscillates / moves to and fro (and collides with molecules)
2 compressions and rarefactions / molecules vibrate / longitudinal wave | B1
R1 |

May/June 5054/22

Question 10 · included (b)

- (b) A microphone is connected to a c.r.o. to display a sound wave.

Fig. 10.2 shows the trace on the c.r.o.

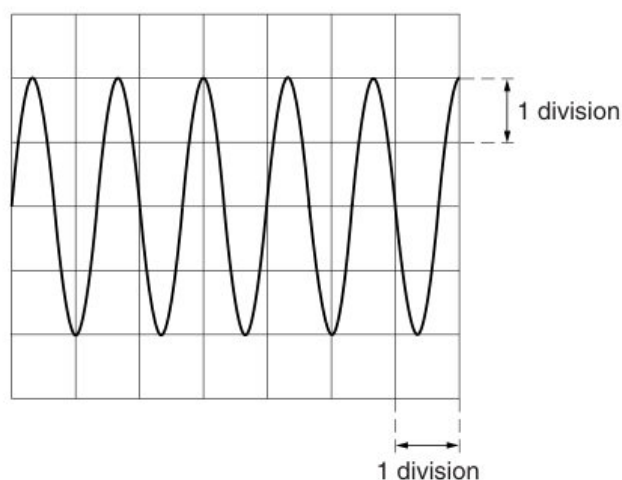


Fig. 10.2

The settings on the c.r.o. are: Y-gain 0.5 V/division; time base 2.0 ms/division.

Mark scheme — only (b)

(b)	(E =) mL numerical or algebraic or 52 000–6000 or 46 000 (J) seen	C1
	(52 000–6000)/20 or 46 000/20 2300 J/g or 2.3×10^6 J/kg	C1 A1
(c)	fast moving/energetic molecules escape/evaporate/break bonds/become gas leaving slow(er) molecules/less energetic molecules or reducing average (kinetic) energy (of molecules or liquid)	B1 B1
5	(a) (i) long sight or far sight or hypermetropic	B1
	(b) (i) 1 1(.0)V	B1
	2 one wave 1.3–1.4 squares or 3 waves in 4 squares 2.6–2.8 ms	C1 A1
	3 (f =) 1/T numerical or algebraic 345–400 Hz	C1 A1
	(ii) smaller amplitude shown larger period shown	B1 B1
	(b) (i) 2	B1
	(ii) neutron	B1
	(iii) 2	B1
	(iv) 4	B1
(c)	nuclei repel or like/positive charges repel (needs) high <u>kinetic</u> energy/speed (to overcome repulsion)	B1 B1

Question 6 · included (a), (b)

EM colours

- (a) State which colour in Fig. 6.1 has

- (b) Visible light, infra-red, ultra-violet and radio waves are four components of the electromagnetic spectrum.

State two other components of the electromagnetic spectrum and describe a different use that is made of each component.

1. component

description of use

.....

.....

2. component

description of use

.....

.....

[4]

Mark scheme — only (a), (b)

6 (a) (i) red B1

(ii) blue B1

(b) ANY 2 from (the use must agree with the type) B4

Microwaves B1

use – satellite television, telephone, mobile / cell phones; B1

cooking, heating in a microwave oven, television

remote, radar, communication

X(-rays) B1

use – hospital use in medical imaging or security imaging, killing cancerous cells, B1

and engineering applications such as detecting cracks in metal. crystallography

May/June 5054/21

Question 7 · included (c)

- (c) The loudspeaker produces sound of frequency 500Hz. The speed of sound is 320m/s. Calculate the wavelength of the sound produced.

wavelength =[2]

Mark scheme — only (c)

Use official MS for the listed parts only.

Question 5 · included (a), (b)

- 5 (a) Light is a transverse wave. The direction of vibration is perpendicular to the direction of transfer of the energy.

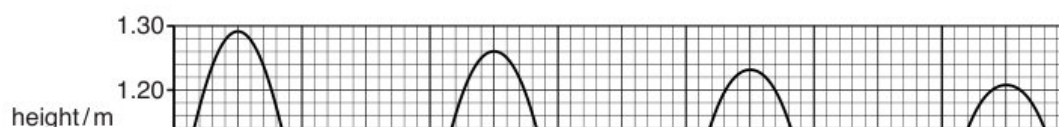
Complete the table of Fig. 5.1 to show the direction of vibration and the type of wave associated with a sound wave and with a wave on the surface of water in a ripple tank.

wave	direction of vibration	type of wave
light	perpendicular to the direction of transfer of the energy	transverse
sound		
water		

[2]

Fig. 5.1

- (b) A wave travels along a rope. Fig. 5.2 shows how the height of one point on the rope varies with time as the wave passes.



Mark scheme — only (a), (b)

- 5 (a) *sound*: along or parallel (to transfer of energy or wave) **and** longitudinal B1
water: perpendicular **and** transverse B1
- (b) (i) 0.29 – 0.28 m B1
- (ii) time / period for one wave(length) / cycle constant B1
or each oscillation / cycle takes one second

2015

Oct/Nov 5054/21

Question 7 · included (a), (b)

- (a) Describe one difference between a compression and a rarefaction.

.....

 [1]

- (b) In a particular medium, the speed of both audible sound and ultrasound waves is v .

Mark scheme — only (a), (b)

7	(a)	at compression: molecules closer together or pressure higher or vice versa for rarefaction	B1	
	(b)	(i) $v = f\lambda$ or in words	B1	
		(ii) larger and because the frequency is lower	B1	
	(c)	states one use basic idea	(e.g. prenatal scanning) (e.g. ultrasound reflects off foetus)	B1 B1 [5]

May/June 5054/22

Question 9 · included (a), (b), (c), (d)

(a) Explain what is meant by the *frequency* of a wave.

.....

 [2]

(b) (i) Determine the wavelength of the wave in deep water.

..... [1]

(c) The wave passes from deep water into shallow water. The speed of the wave is less in shallower water.

(i) State and explain how this affects the wavelength of the wave.

.....

 [2]

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(d) Sound is also a wave.

Mark scheme — only (a), (b), (c), (d)

9	(a)	number of waves (that pass a point) or number of oscillations (passing a point) in unit time or per second or in 1 second	M1 A1
	(b)	(i) 1.5 cm	B1
		(ii) $(v =)f\lambda$ or 5×1.5 seen or $(s=)d/t$ and $f = 1/t$ 7.5 cm/s	C1 A1
	(c)	(i) wavelength decreases travels a shorter distance in the same time or frequency stays the same (and $v = f\lambda$)	B1 B1

May/June 5054/21

Question 4 · included (a), (b), (c)

(a) Describe the movement of the particles of the string.

.....

.....

.....

.....[2]

(b) Determine the wavelength of the wave.

wavelength =[1]

(c) The speed of the wave is 2.0 m/s.

Mark scheme — only (a), (b), (c)

4	(a)	oscillate/vibrate stated or described transverse movement described	B1
			B1
	(b)	0.40 m	B1
(c)	(i)	$v = f\lambda$ or $(f =) v/\lambda$ or $2/(b)$ 5.0 Hz	C1 A1
	(ii)	clear attempt to draw wave moved along 0.20 m to right	B1